Government of Tamilnadu

Department of Employment and Training

Course : TNPSC Group I Mains Material

Subject : Role and impact of Science and Technology in the development of India

Topic : Achievements of Indians in the fields of Science & Technology

© Copyright

The Department of Employment and Training has prepared the TNPSC Group-I Preliminary and Main Exam study material in the form of e-content for the benefit of Competitive Exam aspirants and it is being uploaded in this Virtual Learning Portal. This e-content study material is the sole property of the Department of Employment and Training. No one (either an individual or an institution) is allowed to make copy or reproduce the matter in any form. The trespassers will be prosecuted under the Indian Copyright Act.

It is a cost-free service provided to the job seekers who are preparing for the Competitive Exams.

Commissioner,

Department of Employment and Training.

ACHIEVEMENTS OF INDIANS IN THE FIELDS OF SCIENCE AND TECHNOLOGY

Subrahmanyam Chandrasekhar (1910-1995)

S. Chandrasekhar was an Indian American Astrophysicist. He along William A. Fowler won the 1983 Nobel Prize for Physics for key discoveries that led to the currently accepted theory on the later evolutionary stages of massive stars. He is best known for "Chandrasekhar Limit".

Chandrasekhar limit:

The Chandrasekhar limit is the mass above which electron degeneracy pressure in the star's core is insufficient to balance the star's gravitational self-attraction. Consequently, a white dwarf with a mass greater than the limit is subject to further gravitational collapse, evolving into a different type of stellar remnants, such as a neutron star or black hole. Those with masses under the limit remain stable as white dwarfs.

Chandrasekhar determined what is known as the Chandrasekhar limit – that a star having a mass more than 1.44 times that of the sun does not form a white dwarf but instead continues to collapse, blows off its gaseous envelope in a supernova explosion, and becomes a neutron star.

The significance of Chandrasekhar limit:

1. The limit of a star is characterized by thermonuclear fusion; hydrogen fuses to helium, helium to carbon, and so on, creating heavier and heavier elements. However, thermonuclear fusion cannot create elements heavier than iron. Only a supernova explosion can create copper, silver, gold, and the "trace elements" that are important for the processes of life.
2. Lighter elements like carbon, oxygen, and nitrogen are also essential to life, but without supernova explosions, they would remain forever locked up in stars.

3. The Chandrasekhar Limit is therefore not just as an upper to the maximum mass of an ideal white dwarf, but also a threshold. A star surpassing this threshold no longer hoards its precious cargo of heavy elements. Instead, it delivers them to the universe at large in a supernova that marks its own death but makes it possible for living beings to exist.

The Chandra X-ray Observatory was launched and deployed by Space Shuttle Columbia on 23 July 1999. The Chandrasekhar number, an important dimensionless number magneto-hydrodynamics, is named after him. The asteroid 1958 Chandra is also named after Chandrasekhar. In India, the Himalayan Chandra Telescope is named after him.

Satyendra Nath Bose (1894 – 1974)

Satyendra Nath Bose, Indian mathematician and physicist, noted for his collaboration with Albert Einstein in developing a theory regarding the gas like qualities of electromagnetic radiation.

S.N. Bose research's in the 1920s, laid the foundation for quantum statistics and collaborated with Einstein as an equal partner on ground breaking research advances. Before his working relationship with Einstein, Satyendra Nath wrote a short but significant scientific article, "Planck's Law and the Hypothesis of Light Quanta", which introduced the quantum statistics of photons (units of light).

He sent the article to Einstein after it was initially rejected for publication elsewhere. The article was well received. Einstein later adopted ideas of the article and extended them to atoms – a development that would later lead to two seminal concepts in physics: the Bose-Einstein condensate phenomenon (the transition that certain gases in particle physics undergo when cooled), and Bose-Einstein statistics.

Quantum mechanics (including quantum field theory) is a fundamental theory in physics which describes nature at the smallest scales of energy levels of atoms and subatomic particles.

Bose-Einstein Statistics:

Bose-Einstein statistics (B-E statistics) is one of two possible ways in which a collection of non-interacting indistinguishable particles may occupy a set of available discrete energy states, at thermodynamic equilibrium. Particles obeying Bose-Einstein statistics, accounts for the cohesive streaming of laser light and the frictionless creeping of superfluid helium. Such particles have integer values of spin and are named bosons.

Satyendranath's paper also laid the basis for describing the two fundamental classes of sub-atomic particles – bosons (named in honor of his last name – Bose), and fermions, named after the Italian physicist Enrico Fermi

Bonsons are subatomic particle with integral spin (i.e., angular momentum in quantum-mechanical units of 0, 1, etc.) that is governed by the Bose-Einstein statistics Bosons include mesons, nuclei of even mass number (e.g., helium-4), and the particles required to embody the fields of quantum field theory.

Higgs boson (or "God particle") is today the focus of research at the Large Hadron Collider particle accelerator located at the European Organization for Nuclear Research (CERN). As testament to the foundations laid by Satyendranath's early work, the Higgs Boson is the standard model of the particle physics theory that is believed to provide clues to the Big Bang, and insight into why matter has mass

Bose-Einstein Condensate:

Bose-Einstein condensate (BEC) is what happens to a dilute gas when it is made very cold, near absolute zero, i.e. $0^{o}K$ (which equals – 273 ^{o}C or -459.67^{o}F). It forms when the particles that make up have very low energy. Only bosons can make a Bose-Einstein condensate. The gas has extreme low density, about one-hundred thousandth the density of normal air.

Application of B-E Condensate:

1. BEC, contribute to statistical mechanics, electromagnetic properties of the ionosphere, theories of X-ray crystallography thermoluminescence, and unified field theory.

2. Quantum information processing-the concept of a quantum computer.
3. Precision measurement by the development of most sensitive detectors using BEC
4. Development of optical lattices which could be easily modifiable by varying the interplanar spacing etc.

G.N. Ramachandran (1922-2001)

Gopalasamudram Narayanan Ramachandran, (GNR) was an Indian physicist who known for his work of the Ramachandran plot for understanding peptide structure. He was the first to propose a triple-helical model for the structure of collagen.

Using X-ray diffraction Ramachandran along with Gopinath Kartha proposed and published the triple helical structure of collagen in 1954 in the journal Nature drawing worldwide scientific attention to the "Madras group".

Collagen Helix:

Collagen is the main structural protein in the extracellular space in the various connective tissues in animal bodies. As the main component of connective tissue, it is the most abundant protein in mammals, making 25% to 35% of the whole-body protein content. Collagen consists of amino acids wound together to form triple-helices to form of elongated fibrils. It is mostly found in fibrous tissues such as tendons, ligaments and skin, Collagen helix, or type-2 helix, is a major shape in secondary structure. It consists of a triple helix made of the repetitious amino acid sequence glycine X-Y, where X and Y are frequently proline or hydroxyproline.

The significance of the Ramachandran Plot

GNR's undying quest for the study of collagen led to the historical Ramachandran Plot in 1963. Researchers use the Ramachandran Plot to study protein structures even today.

Ramachandran can be credited for bringing together into one field of molecular biophysics the then disparate fields of X-ray crystallography, peptide synthesis, NMR and other optical studies, and physicochemical experimentation.

In 1999, the International Union of Crystallography honoured him with the Ewald Prize for his 'outstanding contributions to crystallography'. Each year, the Council of Scientific and Industrial Research (CSIR) awards the 'G.N. Ramachandran Gold Medal for Excellence in Biological Sciences & Technology" in his memory for work in Biological Sciences & Technology.

Meghnad Saha:

Meghnad Saha was a great Indian scientist who made a remarkable contribution to the field of astrophysics. He put forward an ionization formula which explains the presence of the spectral lines.

He belonged to a low-income family and struggled to rise in life. He was born in Bangladesh on 06 October 1893. In 1917, he started his professional career and joined a lecturer at the newly opened university college of Science in Calcutta. He taught quantum physics. Along with S.N. Bose, he translated the papers published in German by Albert Einstein and Hermann Minkowski on relatively into the English version. In 1919, the American Astrophysical journal published. On selective radiation pressure and its, application, a research paper by Meghnad Saha. He put forward an ionization formula which explains the presence of spectral lines which proved out to be a breakthrough in Astrophysics and he went abroad for further research. Later in his career, he got interested in Nuclear Physics. In 1950, India had its first cyclotron in operation which was installed due to efforts of Saha. He invented an instrument to measure the weight and pressure of solar rays. He produced the famous equation which he called 'equation of the reaction-isobar for ionization" which later became known as Saha's 'Thermo-ionization equation'. He was also the Chief Architect of river planning in India. He prepared the original plan for the Damodar valley project. He was nominated for the noble prize in physics in 1935-36 in 1952, he was elected a Member of Parliament from West Bengal.

Contributions:

Saha worked on diverse topics as reflected in the titles of his published research papers as indicated below:

1. On Maxwell's Stresses. This paper was based on his studies of the electromagnetic theory of radiation.
2. On the limit of interference in the Fabry -Perot Interferometer
3. On a new theorem in elasticity
4. On the dynamics of electron
5. On the pressure of the light
6. On the influence of finite volume of molecules on the equation of state. This paper was jointly written with SN Bose.
7. On the mechanical and electrodynamical properties of the electron.
8. On the radiation pressure and the quantum theory.

Sir Chandrasekhara Venkata Raman

RAMAN (born November 7, 1888, Trichinopoly, India-died November 21, 1970, Bangalore), an Indian physicist whose work was influential in the growth of science in India.

National Science Day is celebrated in India on 28 February each year to mark the discovery of the Raman Effect by Indian physicist Sir C.V. Raman on 28 February 1928.

Awards & Achievements

1. He won the 1930 Nobel Prize in Physics "for his work on the scattering of light and the discovery of the Raman Effect", becoming the first Indian to win a Nobel Prize in the sciences.
2. He was honoured with the Bharat Ratna, India's highest civilian award, in 1954 in recognition his invaluable contributions to the field of science.
3. Raman was also awarded the 'Lenin Peace Prize' in 1957.

His major contributions

1. **Raman Effect:** After intensive study at his laboratory in Bangalore, he concluded that when a light beam travels through a medium, the beam is deflected by the molecules. But more importantly, he observed that a small part of the emerging light beam after deflection by the molecules had a different wavelength from the original beam. This change in wavelength of the light is known as the Raman Effect.

2. **The Colour of the Sea:** With the help of this phenomenon he also discovered the reason why the colour of ocean water appears blue. The reason was that of scattering of light from the water. Before Raman Introduced his theory, it was believed that the sea gets blue colour from the sky of due to suspended matter in the water.
3. **Raman Spectroscopy:** Raman showed that the energy of photons scattered inelastically serves as a 'fingerprint' for the substance the light is scattered from. As a result of this, Raman spectroscope is now commonly used in chemical laboratories all over the world to identify substances. It is also used in medicine to investigate living cells and tissues – even detecting cancers – without causing harm, Laser light rather than sunlight is used as the source of photons.

Har Gobind Khurana (1922 – 2011)

Har Gobind Khorana, Indian-born American biochemist, an organic chemist who specialized in the study of proteins and nucleic acids.

Awarded Nobel Prize for:

Khorana's work uncovered how a DNA's genetic code determines protein synthesis – which dictates how a cell function. That discovery earned Khorana, along with two colleagues, the 1968 Nobel Prize in Physiology or medicine "for interpretation of the genetic code and its function in protein synthesis". Hargobind Khorana shared the Nobel Prize in Physiology of Medicine with Robert W. Holley and Marshall W. Nirenberg in 1968.

Khorana was the first scientist to chemically synthesize oligonucleotides. This achievement, in the 1970s, was also the world's first synthetic gene; in later years, the process has become widespread. Subsequent scientists referred to his research while advancing genome editing with the CRISPR/Case system.

Oligonucleotides are short DNA or RNA molecules, oligomers, that have a wide range of applications in genetic testing, research, and forensics.

These small bits of nucleic acids can be manufactured as single-stranded molecules with any user-specified sequence, and so are vital for artificial gene synthesis,

polymerase chain reaction (PCR). DNA sequencing, library construction and as molecular probes.

The contribution of Hargobind Khorana:

Developed methods for investigating the structure of the nucleic acids, Khorana introduced many of the techniques that allowed scientists to decipher the genetic code and show how ribonucleic acid (RNA) can specify the structure of proteins.

Created the first synthetic gene – a step that led to commercialized gene synthesis at businesses around the world.

Synthesized the gene for rhodopsin, a protein involved in vision.

Application of H.G. Khorana's Work

These custom-designed pieces of artificial genes are widely used in biology labs for sequencing, cloning and engineering new plants and animals.

These genes are integral to the expanding use of DNA analysis to understand gene-based human disease as well as human evolution.

Srinivasa Ramanujan (1887 – 1920)

Srinivasa Ramanujan was an Indian mathematician who lived during the British Rule in India. Though he had 'almost no formal training. In pure mathematics, he made substantial contributions to mathematical analysis, number theory, infinite series, and continued fractions.

Many of his mathematical discoveries were based on pure intuition – but most of them were later proved to be true. Ramanujan independently compiled nearly 3,900 results (mostly identities and equations).

Ramanujan was the second Indian to be inducted as a Fellow of the Royal Society with some of the most eminent scientists. He joined his fellowship in 1918 at the age of 31, as one of the youngest yellows in the history of society.

Ramanujan's major contribution:

1. Ramanujan prime,
2. The Ramanujan Theta function,
3. The Ramanujan conjecture,
4. Partition formula and
5. Mock theta functions,

I have opened entirely, new areas of work and inspired a vast amount of further research. **Hardy-Ramanujan number 1729:** It is the smallest number that can be expressed as the sum of two positive cube numbers in two different ways.

$$1729 = 1^3 + 12^3 = 9^3 + 10^3$$

Generalizations of this idea have created the notion of 'taxicab numbers' .

Posthumous recognition:

1. His birth anniversary, December 22, is celebrated as the National Mathematics Day every year.
2. There is also a museum dedicated to telling Ramanujan's life story. It is located in Chennai and has many photographs of his home and family.
3. The Man Who Knew Infinity: A Life of the Genius Ramanujan is a biography of Ramanujan, written in 1991 by Robert Kanigel and published by Washington Square Press.

He compiled 3,900 results of great importance before he lost his life at the age of 31. His infinite series for pi was one of his most celebrated findings.

Chintamani Nagesa Ramachandra Rao (1934-)

C.N.R. Rao is an Indian chemist who has worked mainly in solid -state and structural chemistry. He currently serves as the Head of the Scientific Advisory Council to the Prime Minister of India.

Rao is one of the world's foremost solid state and materials chemists. He has contributed to the development of the field over five decades. His work on transition

metal oxides has led to basic understanding of novel phenomena and the relationship between materials properties and the structural chemistry of these materials.

Major research areas of C.N.R. Rao:

1. Transition metal oxides and other extended inorganic solids (new synthesis and novel structures metal-insulator transitions, CMR materials, superconductivity, multiferroics etc.),
2. Inorganic-organic hybrid materials,
3. Nanomaterials including nanotubes and graphene,
4. Artificial photosynthesis and hydrogen generation by photo catalysis and thermal means.

CNR Rao received the Bharat Ratna Award in 2014, making him the 3rd scientist after C.V. Raman and A.P.J. Abdul Kalam to receive the award.

Tessy Thomas:

Tessy Thomas is an Indian scientist and Director General of Aeronautical Systems and the former Project Director for Agni-IV missile in Defence Research and Development Organisation. She is the first woman scientist to head a missile project in India. She is known as the 'Missile Woman' of India.

The first woman to head an Indian missile project, Tessy Thomas has decisively broken the glass ceiling to make her mark in a traditionally male bastion. Having stood out ever since she joined the DRDO in 1988, she has played a pivotal role in India's missile development programme, particularly in the making of its long-range nuclear-capable ballistic missile, the Agni-V.

The contribution of Tessy Thomas:

1. Tessy was associate project director of the 3,000 km range Agni-ill missile project.
2. She was the project director for mission Agni IV-which was successfully tested in 2011.
3. Tessy was appointed as the Project Director for 5,000 km range Agni-V in 2009. The missile was successfully tested on 19 April 2012.

She worked under her revered role model, APJ Abdul Kalam, who placed her in the Agni missile programme. She was appointed as Director-General, Aeronautical Systems of DRDO in 2018. She is the recipient of Doctor of Science (Honoris causa) from five different Universities; this brilliant woman is also a fellow of various professional institutes – Indian National, Academy of Engineering (NAE) Institution of Engineers-India (IEI), Tata Administrative Service (TAS).

A.P.J. Abdul Kalam (1931-2015)

APJ Abdul Kalam was an Indian politician and aerospace scientist who served as the 11th President of India from 2002 to 2007. He worked extensively with Defense Research and Development Organisation (DRDO) and Indian Space Research Organization (ISRO) and was intimately involved in India's civilian space programme and military missile development efforts. He thus came to be known as the Missile Man of India for his work on the development of ballistic missile and launch vehicle technology.

Kalam promoted the advancement of the national nuclear weapons program. Kalam also devised a 20-year action plan (Vision 2020) to achieve economic growth through technological development in India.

Contribution in DRDO:

1. He played a pivotal organizational, technical, and political role in India's Pokhran-II nuclear tests in 1996, the first since the original nuclear test by India in 1974.
2. He was chief executive of integrated Guided Missile Development Programme (IGMDP), where he played an important role developing many missiles under the mission including Agni, an intermediate range ballistic missile and Prithvi, the tactical surface-to-surface missile.

Role in ISRO:

He was the project director of India's first Satellite Launch Vehicle (SLV-III) which successfully deployed the Rohini satellite in near-earth orbit in July 1980.

Other Contribution:

In 1998, along with cardiologist Soma Raju, Kalam developed a low-cost coronary stent, named the “Kalam-Raju Stent”.

In 2012, the duo designed a rugged tablet computer for health care in rural areas, which was named the “Kalam-Raju Tablet”.

In 1998, Kalam put forward a countrywide plan called Technology Vision 2020, which he described as a road map for transforming India from a less-developed to a developed society in 20 years. The plan called for, among other measures, increasing agricultural productivity, emphasizing technology as a vehicle for economic growth, and widening access to health care and education. Among his numerous awards were two of the country’s highest honours, the Padma Vibhushan (1990) and the Bharat Ratna (1997).

Questions:

1. What are the contributions made by Indian Nobel laureates in the field of Science and Technology?
2. Write a note on
   a) Collagen Helix b) Bose-Einstein condensate.
3. What is the significance of Chandrasekhar limit?